



KRONOS

EOS

The dawn of accessible 3D-printed electronics.

Investment case for a 25k-100k € five-axis AME platform
built on the technology of the category's pioneer.

An option-structured bet on the vacant middle of 3D-printed electronics

The ask

€2.5M Tranche 1 of a €13.2M staged plan — committed only gate by gate.

- ✓ Proven 5-axis core technology — acquired at distressed cost from the Neotech insolvency
- ✓ Only vendor able to put true 5-axis AME below every procurement threshold
- ✓ Category leadership vacant: Nano Dimension exited AME in April 2026
- ⚠ Honest framing: today's machine market is small — this is a market-creation bet with capped downside

What the staged €13.2M can build (Base case)

€16.0M

revenue 2031
(Base case)

452

machines installed
by 2031

50%

gross margin 2031,
19% recurring revenue

2031

EBITDA break-even
(Upside: 2030)

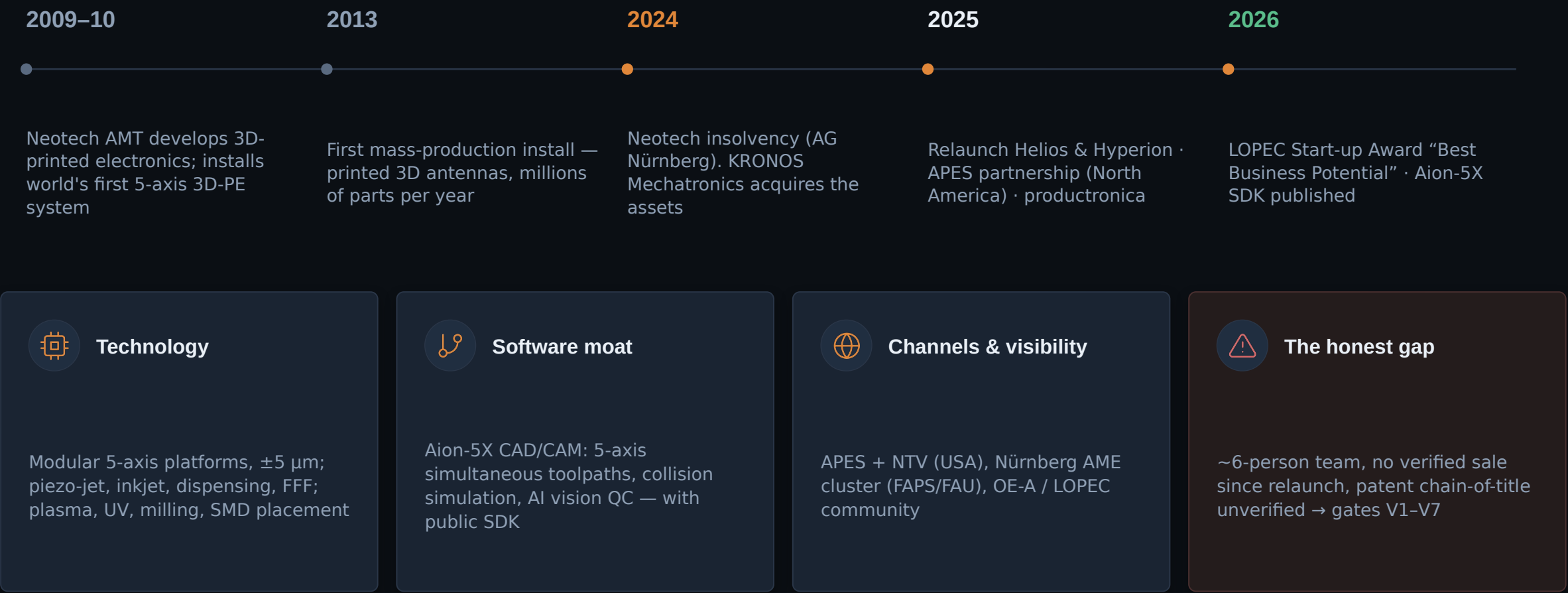
11.4x

MOIC in the Upside path
(IRR ≈ 48%, exit yr 7)

€2.5M

maximum at risk before
series-production gate

The pioneer's technology, reborn as a startup



Brilliant technology, trapped in a €200k+ project business

~100–190

AME machines sold per year, worldwide, across all vendors (bottom-up)

€15–28M

total annual AME machine market today — smaller than one mid-size Mittelstand firm

50–80%

of placements are grant-funded research buys — one-off, non-repeating

2024–2026: the category cleared itself out

Neotech AMT

The 5-axis pioneer — insolvent July 2024 after ~20 years and ~50 systems. Its assets became KRONOS.

Nano Dimension

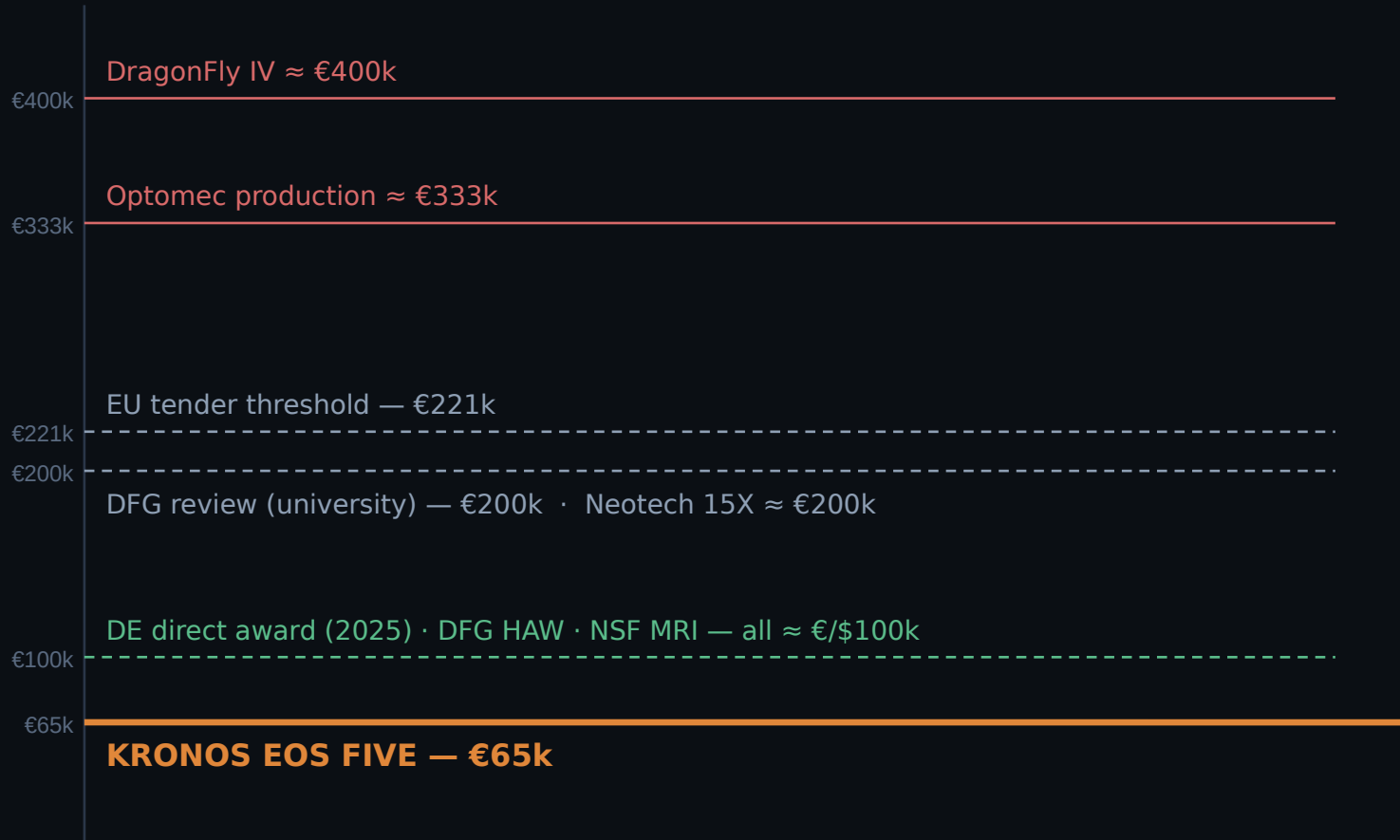
Raised ~\$1.5B. Sold ~51 DragonFly systems.Exited AME in April 2026 for \$2M upfront — the leader left at scrap value.

J.A.M.E.S GmbH

The Hensoldt-backed AME community platform — liquidated August 2025. The ecosystem lost its anchor.

The diagnosis: every player sold €150–500k project machines into a research niche. Nobody productized the technology for the budgets that actually buy lab equipment.

Procurement thresholds decide who can buy — today's prices sit above all of them



Below €100k, a machine purchase changes species

- ✓ No EU tender, no DFG review — a department-level decision
- ✓ Fits a single US faculty startup package (\$300–500k)
- ⚡ German direct-award limit raised to €100k in 2025 — the door just opened
- 🕒 Buying window: weeks, not budget years

The 25–100k € band is empty — for the machines that matter

DESKTOP · <€12k

Voltera V-One · ~\$5k
BotFactory SV2 · ~\$10k
nano3Dprint · ~\$5k

*Planar, education-grade.
5,000+ Voltera units prove the buyer base exists.*

€25–100k · THE GAP

Voltera NOVA · \$46k — planar only
Fujifilm Dimatix · ~\$55k — planar only
LPKF ProtoMat · subtractive milling

*NO five-axis / conformal AME system exists here.
The defining capability of 3D electronics is absent from the only band procurement loves.*

INDUSTRIAL · >€150k

Neotech/KRONOS 15X-class · €200k
XTPL Delta · €170–220k
Optomec, nScript · €150–500k

*Quote-only project machines.
The tier that bankrupted its pioneer.*

← KRONOS EOS →

Machines at this price point sell in the thousands — just not additive 3D ones yet

5,000+

Voltera units shipped to 92 countries — NASA, MIT, Harvard, Stanford, Oxford among customers

The buyer universe is real and global

€26.2M

LPKF Development segment revenue 2024 — “thousands” of ProtoMat benchtop PCB machines installed

EE labs pay 10–60k € for in-house prototyping, for decades

130,000+

Formlabs printers sold after cutting professional SLA prices ~10x — a \$2B company

The 10x price-cut playbook works for professional machines

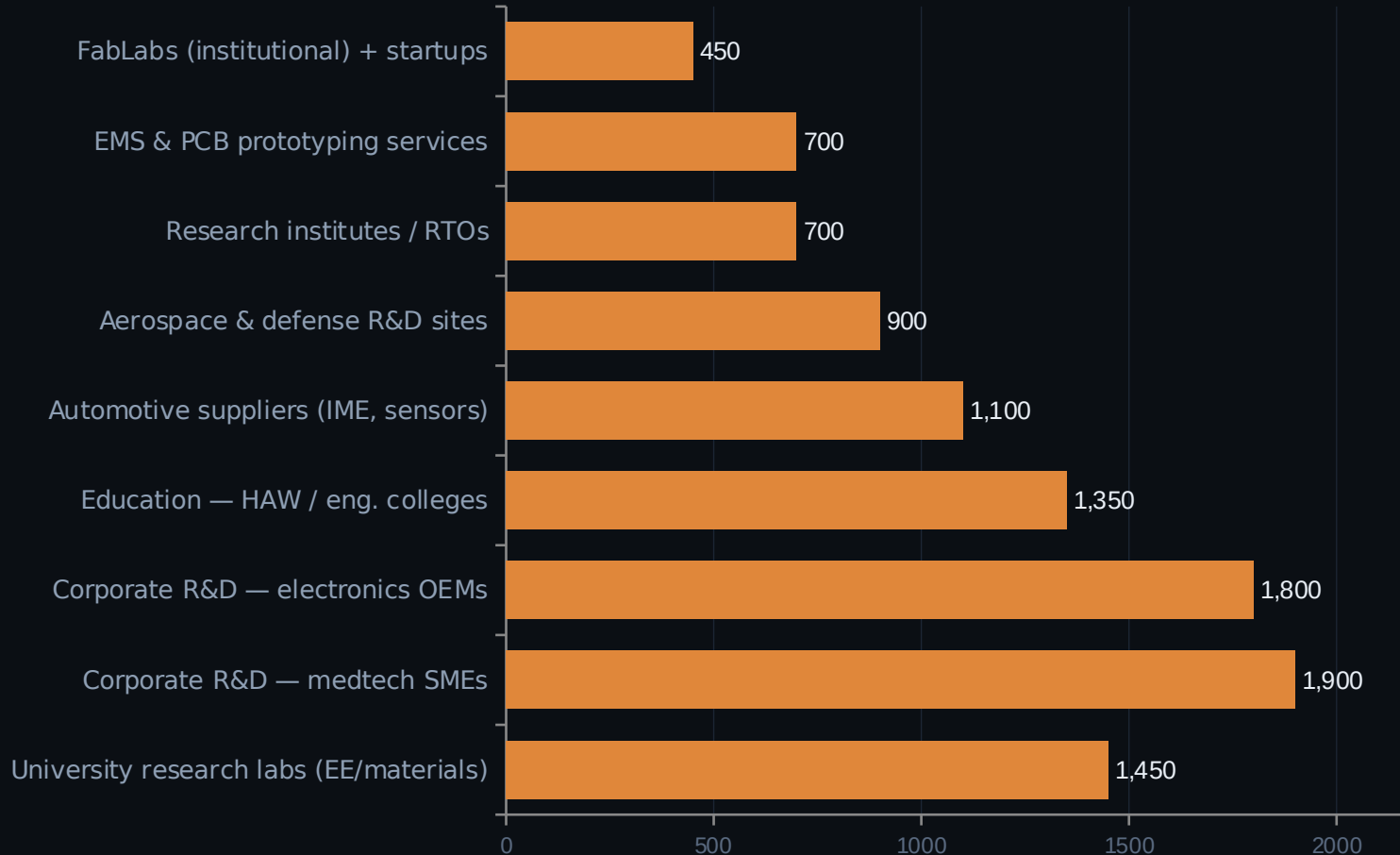
40+ / 13

XTPL Delta systems cumulative / delivered in 2025 alone at €170–220k — 71% research customers

Even at 3x our price, research demand is accelerating

NEW CUSTOMER UNIVERSE

~10,000 organizations that could never buy a €250k machine



Counted bottom-up, not asserted

963 research-active EE universities (Research.com) · 76 Fraunhofer institutes · ZVEI 1,100+ · CLEPA 3,000+ · MedTech Europe 38,000 (90% SME) · in4ma 2,160 EMS · relevance-filtered per segment.

~10,000

candidate organizations — vs. ~150 machines the industry sells per year today

One beachhead, two expansions — sequenced by procurement friction

BEACHHEAD · YEAR 1–2

Research labs & institutes (EU/NA)

- ~2,150 organizations, tender-free below €100k
- Technically tolerant — research forgives v1.0
- Every paper is a free reference (Voltera playbook)
- KRONOS home turf: FAPS cluster, LOPEC award

EXPANSION · YEAR 2–3

Corporate R&D — electronics, automotive/IME, medtech

- Reference-driven entry from beachhead
- IP-sensitive iteration stays in-house
- IME pre-development without tooling cost
- ~4,800 filtered organizations

EXPANSION · YEAR 3–5

Defense & space labs · education fleets · small series

- DoD AM budget: \$0.3B → \$0.8B → ~\$2.6B (2030e)
- Eos Max + traceability via APES/NTV
- Eos One curriculum bundles
- Production cells feed Hyperion upsell

Jobs to be done — what a €65k five-axis machine gets hired for



Publish first

“Print the 3D antenna / conformal sensor the competing group can't.” Five-axis capability is publication-grade differentiation — the strongest academic buying motive.



Iterate same-day

“From CAD to functional 3D circuit before the evening — not a 3-week offshore loop.” Substitutes cap value only for flat standard PCBs; 3D iteration has no substitute.



Keep IP in-house

“Defense, automotive pre-development and medical prototypes never leave the building.” ITAR/IP-sensitive work structurally cannot use board services.



Train the next cohort

“Teach industry-relevant AME on real hardware.” Curriculum seats seed tomorrow's corporate purchases — the LPKF ProtoMat dynamic for 40 years.

Alone in the quadrant that matters



Why the quadrant stays open

- Incumbents: cost structures & cannibalization fear block down-market moves
- Voltera: 12-24 months behind on 5-axis CAM software
- New entrants (HumminK, Scrona, Quantica, XTPL) all target sub-micron / display / semiconductor niches
- **Real threat: a Bambu-style entrant — window ≈ 3-5 years**

KRONOS Eos — one platform, three configurations

EOS ONE

€35k

Teach & enter

- 3+2-axis indexed kinematics
- 1 toolhead (micro-dispense) + UV + FFF
- $\pm 25 \mu\text{m}$ deposition accuracy
- Camera calibration · Aion-5X Lite
- For HAW, colleges, top fab labs

EOS FIVE

€65k

The beachhead product

- TRUE 5-AXIS SIMULTANEOUS printing
- 2 toolheads: piezo-jet + dispensing
- $\pm 15 \mu\text{m}$ · AI vision inspection
- Aion-5X Pro · materials starter kit
- For research groups & corporate R&D

≈ 55% of projected mix

EOS MAX

€95k

Qualify & produce

- 5-axis + toolchanger (up to 4 heads)
- SMD placement module · plasma option
- $\pm 10 \mu\text{m}$ · closed-loop QC
- Traceability package (defense-ready)
- Upsell path toward Hyperion class

Why one platform beats three machines — and why the software is the moat



One chassis, three tiers

Shared 5-axis base, B&R control, vision stack. One spare-parts pool, one CE/UL certification, field-upgradeable tiers. R&D: €3.2M instead of €6–8M for three separate machines.



Aion-5X is the real product

5-axis deposition CAM + calibration + AI inspection took ~15 years to build. It already exists, scales at zero marginal cost — and licenses to third parties (APES Matrix6D).



Curated materials program

One silver ink, one dielectric, two polymers — qualified profiles, subscription kits. Cuts support cost, creates the annuity. “Experimental mode” keeps research freedom.

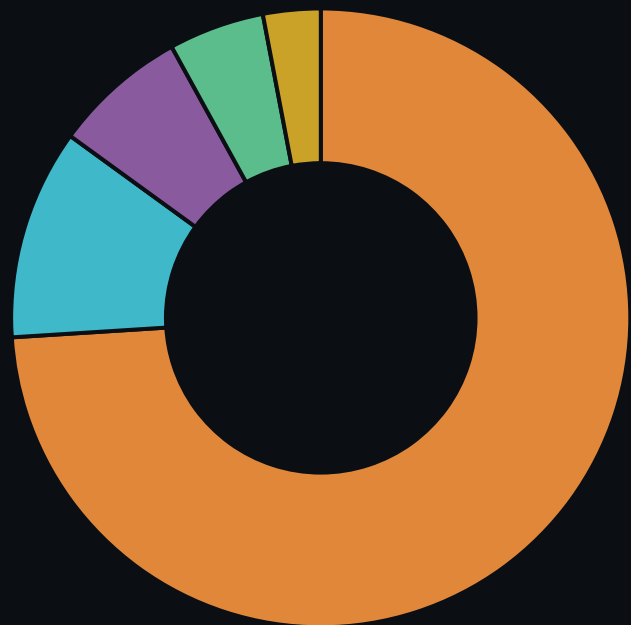


Cannibalization fence

Build volume, throughput, autonomy and material breadth stay with Helios/Hyperion (€200k+). Eos Max is designed as the feeder, not the killer — the Formlabs-Fuse pattern, not MakerBot.

Machines open the account — the installed base pays the margin

Revenue mix 2031 · Base



- Machines 74%
- Consumables 11%
- Services/NRE 7%
- Maintenance 5%
- Software 3%

Per installed machine, per year (Base)

€5.0k

consumables
(curated ink kits)

€3.6k

service contract
(11% · 50% attach)

€1.1k

Aion-5X seat
(60% attach)

≈€10k

recurring per unit
— growing with base

Financing built for SME budgets

Leasing via partners (Grenke/DLL class), balance-sheet neutral: Eos Five ≈ €1.3-1.5k/month — an OPEX decision. German equipment-leasing share: 26.1% of all equipment investment (BDL 2024).

30% down-payment standard keeps working capital at 12% of revenue growth. Machine-as-a-Service tier follows the Carbon precedent (~35-45% of sale price per year).

Sized bottom-up from counted organizations — not from analyst headlines

TAM

≈ €1.2B

10-yr equipment + attach potential · ~10,000 organizations worldwide × 1.3 machines × €65k

SAM

≈ €340M

EU + North America, reachable segments 2027-2031 · ~4,200 organizations · ≈€35-45M/yr by 2031

SOM

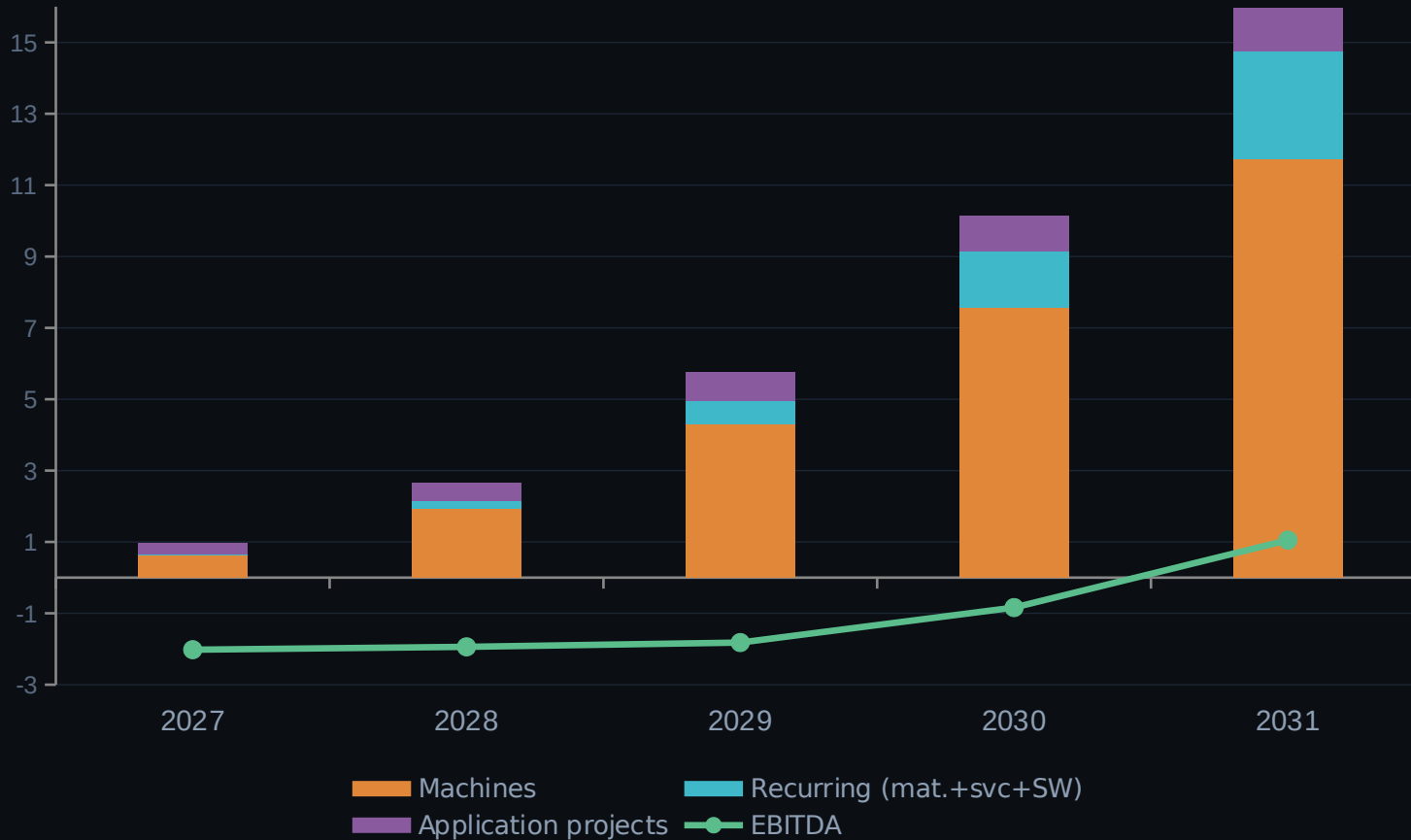
≈ €34M

KRONOS cumulative 5-yr (Base): 452 units + attach ≈ 30% of the matured band

Reality check: today's entire AME machine market is ~€15-28M/yr. Every euro of SAM beyond that assumes the band expands 5-7x by 2031 — the Conservative case assumes zero expansion and is a documented no-go.

5-YEAR FINANCIAL CASE

Base case: €16M revenue, 50% gross margin, EBITDA break-even 2031



Base case anchors

12 → 200

units/yr, 2027 → 2031 (cum. 452)

44% → 50%

gross margin as COGS falls 58% → 48%

€13.2M

gross funding need (peak cash + 15% buffer)

XTPL check

comparable needed €15-18M for 13 units/yr —
Eos starts with finished core tech

Three futures, honestly priced — including the one where you lose

CONSERVATIVE

No band expansion — today's demand only

Units cum. 2031	178
Revenue 2031	€4.9M
EBITDA 2031	−€0.9M
Break-even	none
MOIC / IRR (yr 7)	0.3x / −18%

NOT INVESTABLE — this is the documented no-go scenario the gates protect against.

BASE

Band expands 5–7x — Formlabs-calibrated

Units cum. 2031	452
Revenue 2031	€16.0M
EBITDA 2031	+€1.1M
Break-even	2031
MOIC / IRR (yr 7)	2.25x / +14%

A solid LPKF-class business — below venture threshold on its own.

UPSIDE

Market creation: defense, IME, attach economy

Units cum. 2031	743
Revenue 2031	€29.1M
EBITDA 2031	+€6.8M (23%)
Break-even	2030
MOIC / IRR (yr 7)	11.4x / +48%

Venture-grade. ~25–30% probability carries the deal's expected value (≈3x blended).

Community-led, direct-first — a sales army comes later, if ever

PHASE A · 2027–28

Research beachhead (EU/NA)

- LOPEC · productronica · Formnext presence — award momentum
- Paper-reference program: discount for citability
- Demo tours via FAPS / Fraunhofer network
- Inside sales + 2 application engineers · CAC ≈ €9k

PHASE B · 2028–29

Corporate R&D expansion

- Reference-driven entry: ZVEI / CLEPA / SPECTARIS segments
- Channel activation: APES & NTV (NA), ex-Neotech partners (JP/AU)
- 25% of volume through distributors at 30% discount
- electronica · SMTconnect · IME consortia

PHASE C · 2029–31

Defense · education · small series

- Eos Max + traceability into SBIR / NextFlex ecosystem
- Eos One curriculum bundles for HAW & colleges
- First production-cell clusters — Hyperion feeder
- APAC entry via reactivated distributor network

Six gates, hard kill criteria — €2.5M buys the truth before the factory



TRANCHE 1 — €2.5M

Q4 2026 · funds G0–G4 through pilot fleet

The only commitment requested today

TRANCHE 2 — €4.5M

Q4 2027 · industrialization & launch

Unlocked by: ≥10 LOIs · BOM ≤€26k · IP cleared

TRANCHE 3 — €4–5M

Q1 2029 · scale & channel build-out

Unlocked by: ≥60 units sold · GM ≥42% · service <8%

Substance over optimism

THE BET

For €2.5M, the investor buys the option on the only unclaimed Formlabs position in 3D-printed electronics — proven 5-axis technology, a vacant price band, a cleared field, and procurement rules that just tilted in its favor.



GO — conditionally — Commit T1 after three pre-closing checks: IP chain-of-title, software-team retention, insolvency legacy clean.



STOP at the gates if — <10 LOIs in 6 months · BOM >€32k · NOVA proxy <30 units/yr · service cost >12% in pilots.



DO NOT invest at all if — you require a certain venture return: only the Upside path (~25–30% probability) is venture-grade. This is an option, not a promise.

WHAT THE SKEPTIC SAYS

1

Every predecessor failed or stayed small — maybe the gap is absent demand, not market failure.

2

Zero verified KRONOS sales since relaunch — execution by a ~6-person team is unproven.

3

Academic beachheads are grant-cyclical and don't reorder — the corporate/defense jump must actively succeed.

These are not dismissed — they are the gates. If they hold, the loss is €2.5M and a clean stop. If they fall, the position is category-defining.